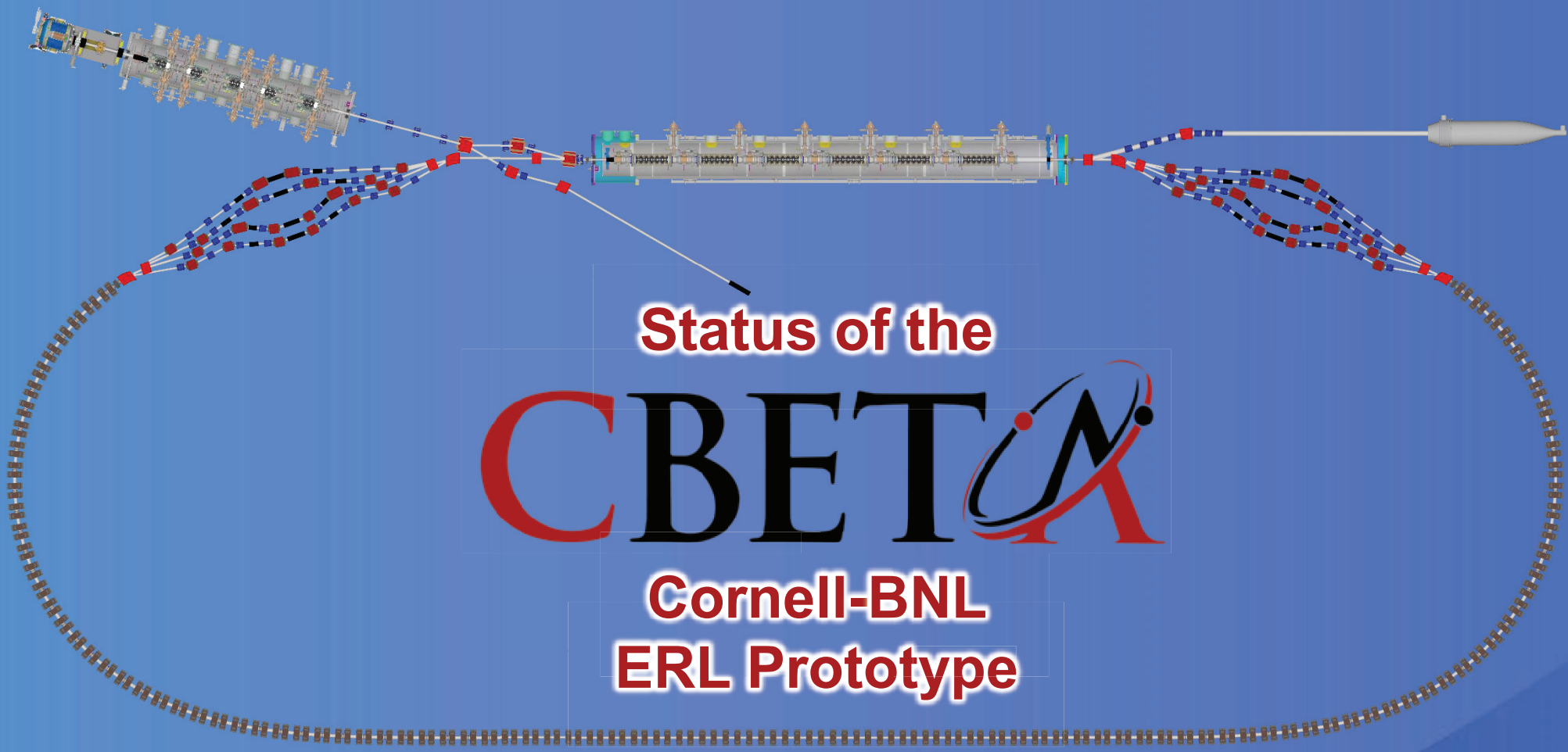




Status of the CBETA Cornell-BNL ERL Prototype

Kirsten Deitrick
On behalf of the CBETA team

BROOKHAVEN
NATIONAL LABORATORY
a passion for discovery

**Office of
Science**
U.S. DEPARTMENT OF ENERGY



Cornell Laboratory for
Accelerator-based Sciences and
Education (CLASSE)

NAPAC2019



- Introduction
- Milestones
- Future Plans
- Conclusion
- Acknowledgments
- References



Background



- CBETA stands for Cornell-BNL ERL Test Accelerator



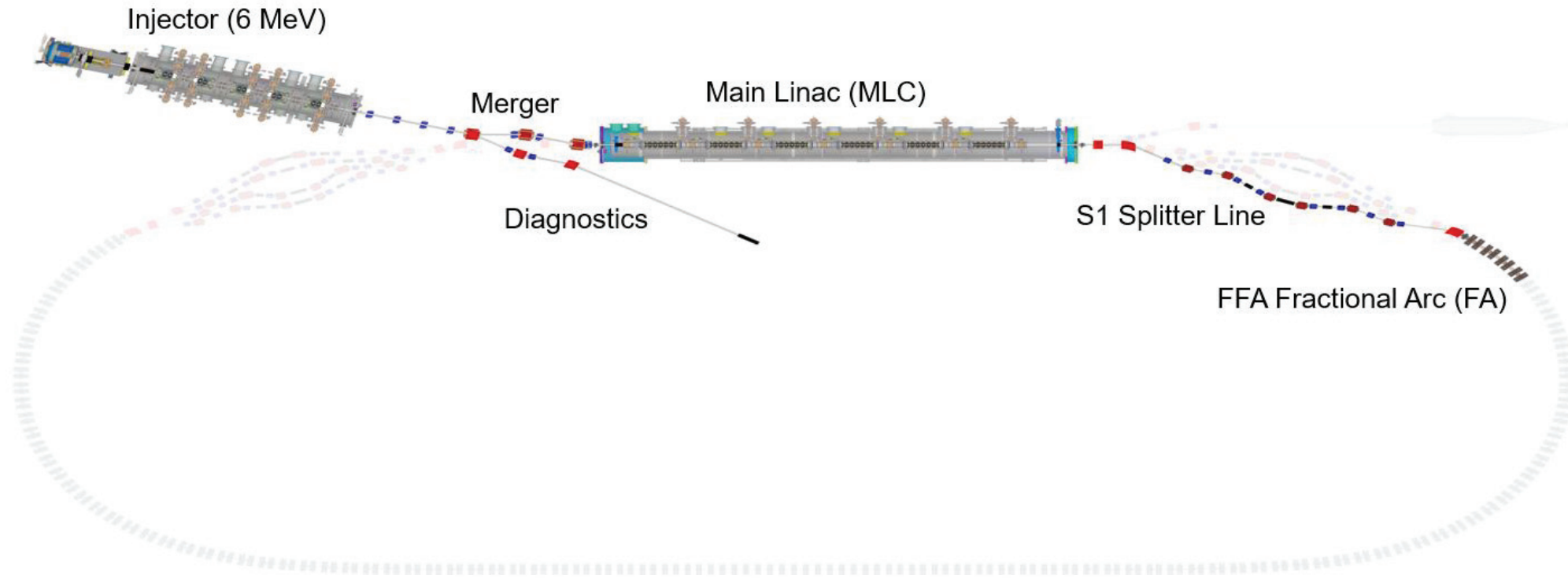
- CBETA stands for Cornell-BNL ERL Test Accelerator
- Multi-pass SRF Energy Recovery Linac (ERL) utilizing a Non-Scaling Fixed Field Alternating-gradient (NS-FFA) permanent magnet return loop, up to 4 passes and 150 MeV
 - FFA section has a wide energy acceptance – all 4 energies in the same pipe



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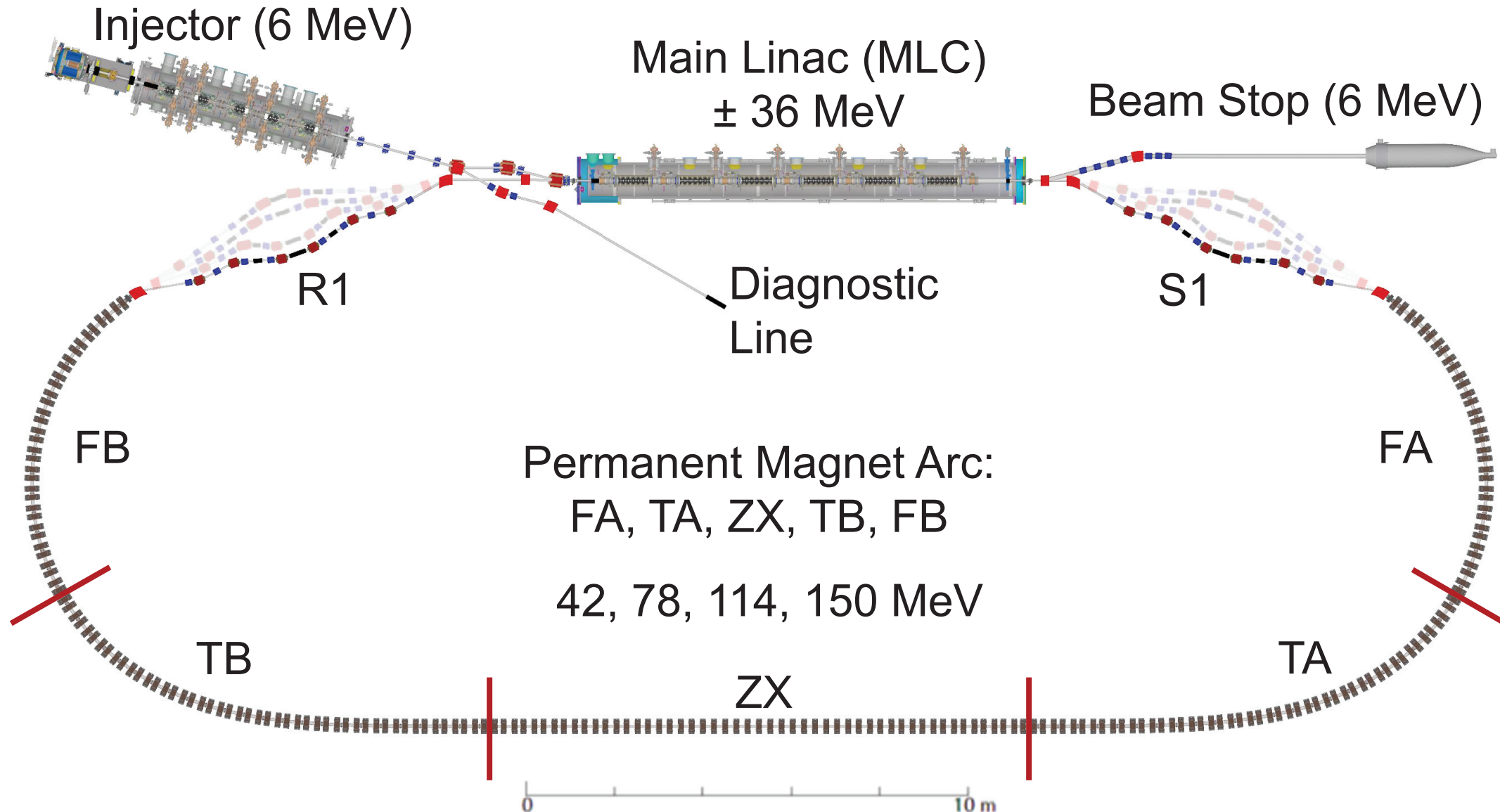
- Fractional Arc Test (FAT) involved the Injector, MLC, S1, and one FFA girder; took place in the spring of 2018



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- Following the successful conclusion of the FAT, the rest of CBETA was approved for completion

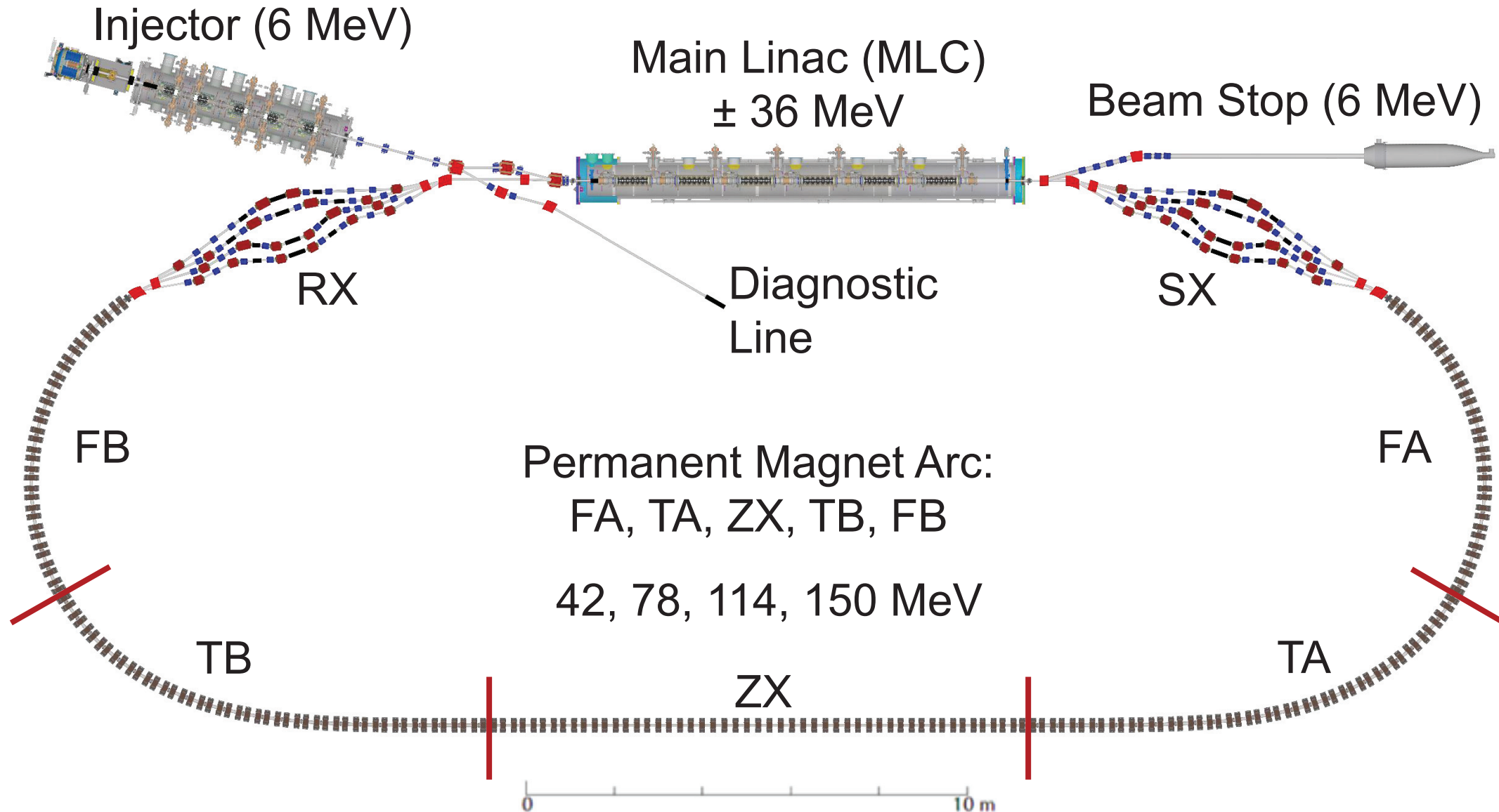


One-Pass Layout

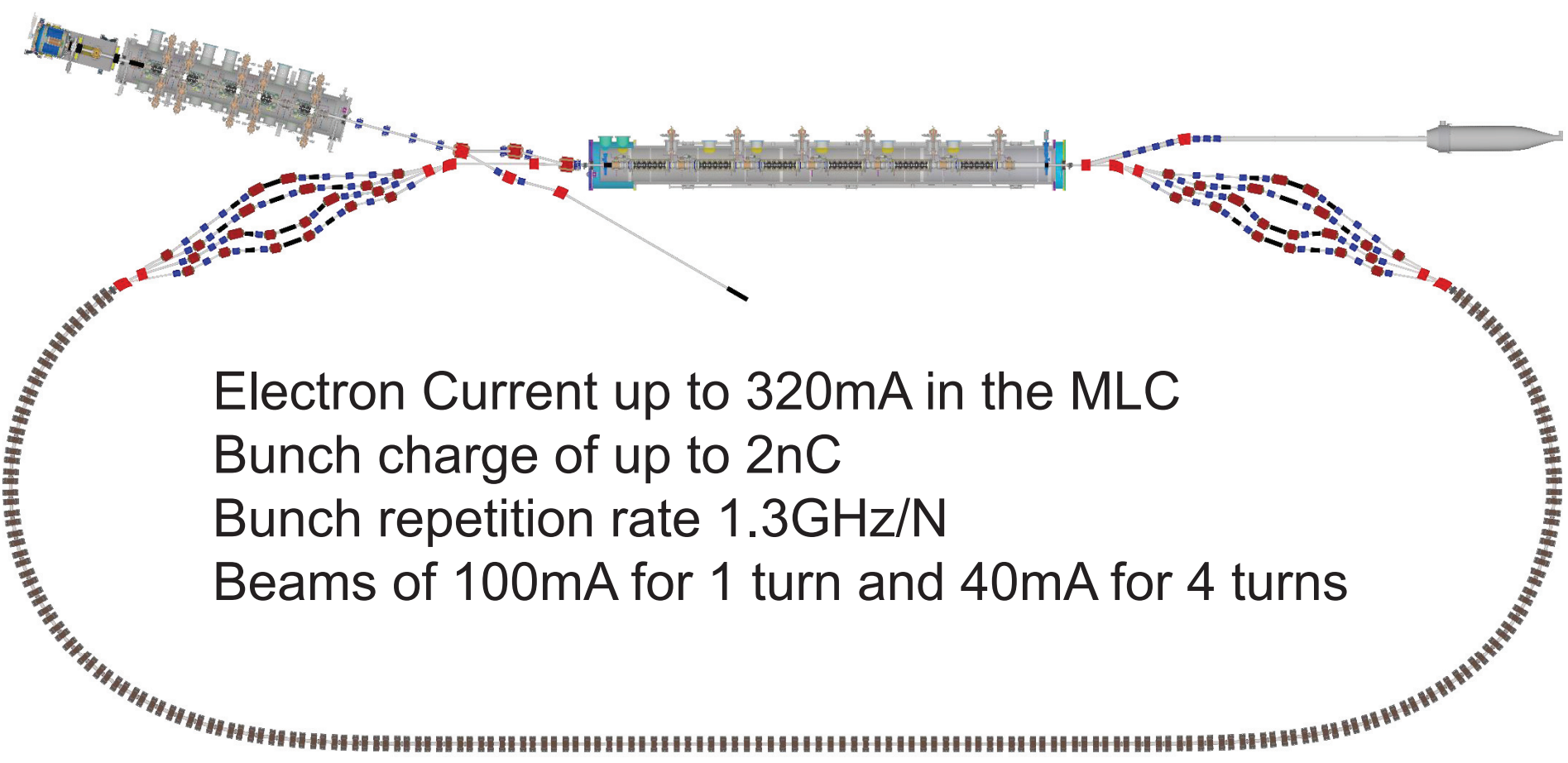




Four-Pass Layout



Design Parameters



Electron Current up to 320mA in the MLC
Bunch charge of up to 2nC
Bunch repetition rate 1.3GHz/N
Beams of 100mA for 1 turn and 40mA for 4 turns

- Cornell DC gun
- 100mA, 6MeV
SRF injector (ICM)
- 600kW beam stop
- 100mA, 6-cavity
SRF CW Linac
(MLC)
- Sliding joints in
each SX/RX line



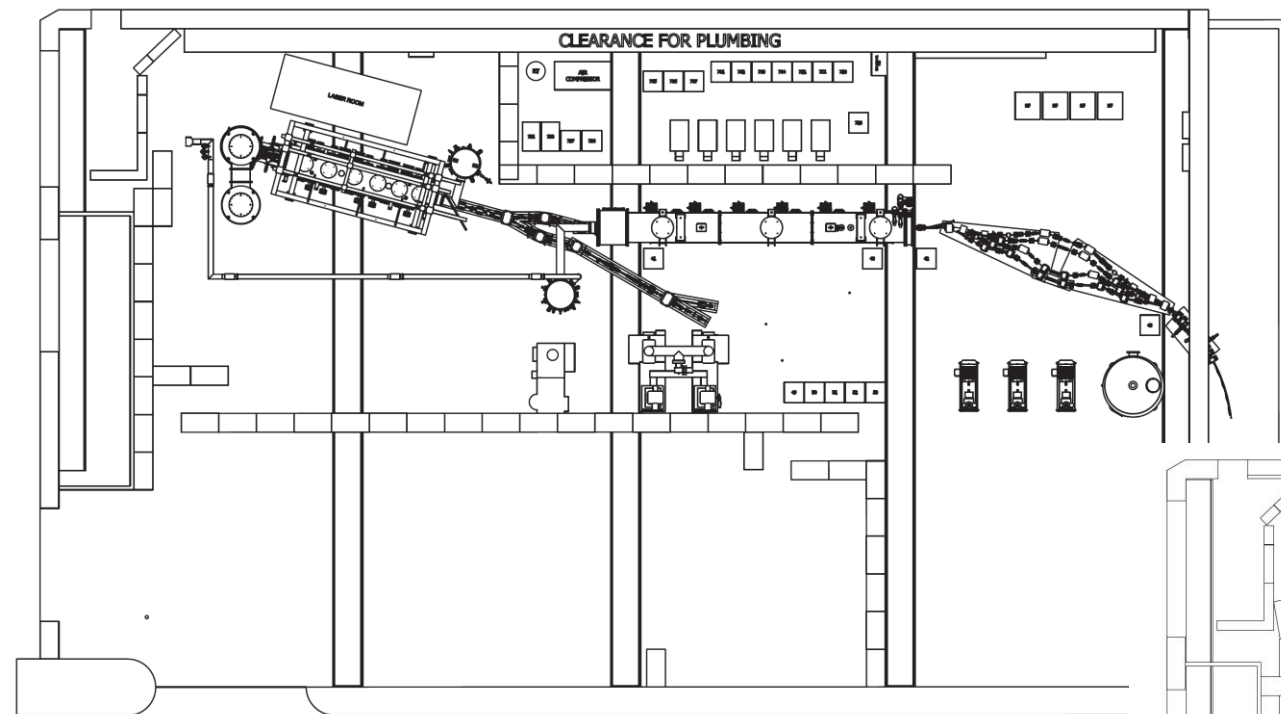
Spring 2018



- CBETA Fractional Arc Test picture, with many of the team members
- Showing S1 line (center) and test FFA girder (top center), with the end of the MLC (bottom right)



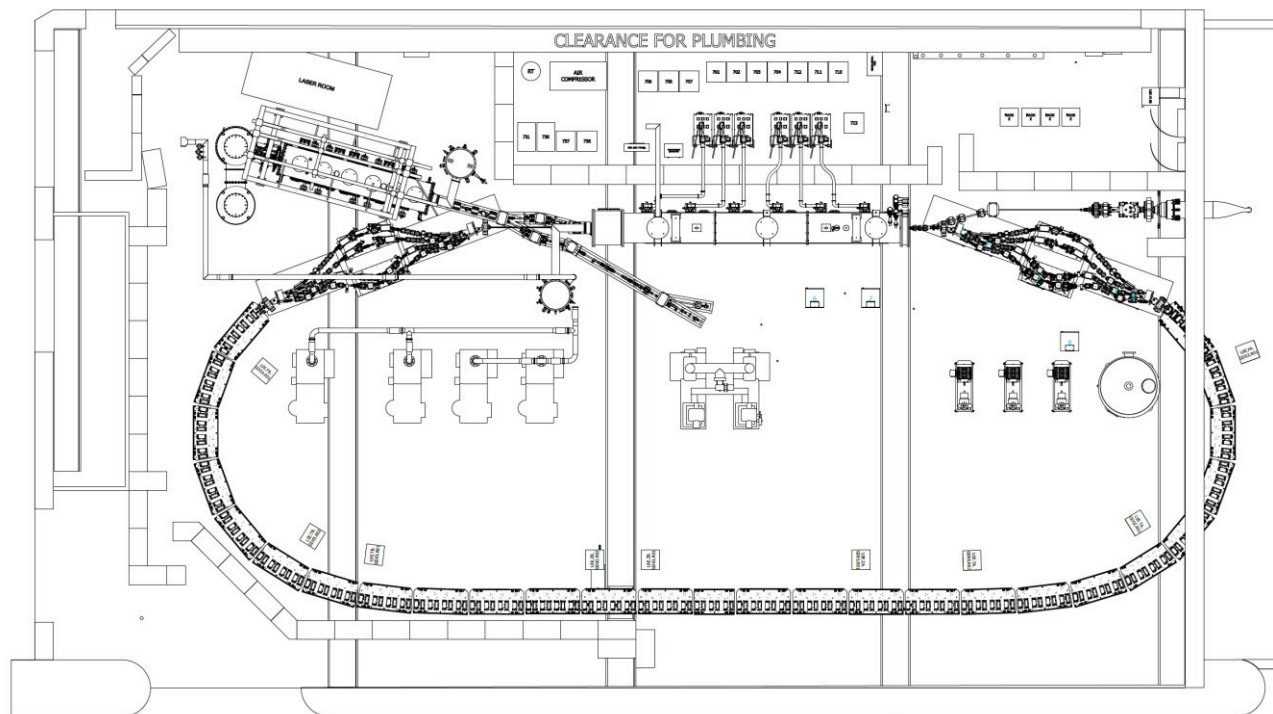
Construction



FAT layout



North (approximately)



CBETA 4-pass layout



Injector and MLC already
installed along the north wall,
MLC shown amidst
construction at the left



Construction

October 2018



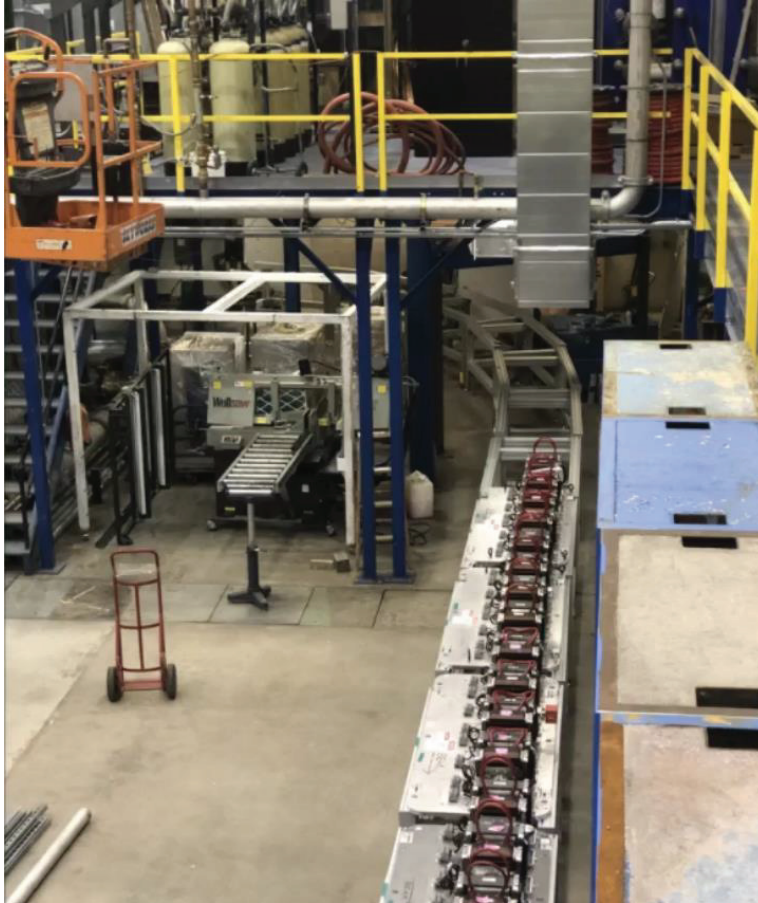
- South side of accelerator enclosure
- Future location of FFA arc
- Prior elements constructed and installed (Injector, MLC, Diagnostic Line) to the left/bottom-left of picture



Construction



January 2019



- South side of enclosure looking east (left) and west (right)
- Girders for FFA arc installed
- FFA magnets on table tops begin installation (assembled at BNL)



Construction



February 2019

- South side looking west (left) and west FFA (bottom)
- FFA magnets installed, cooling water lines installed

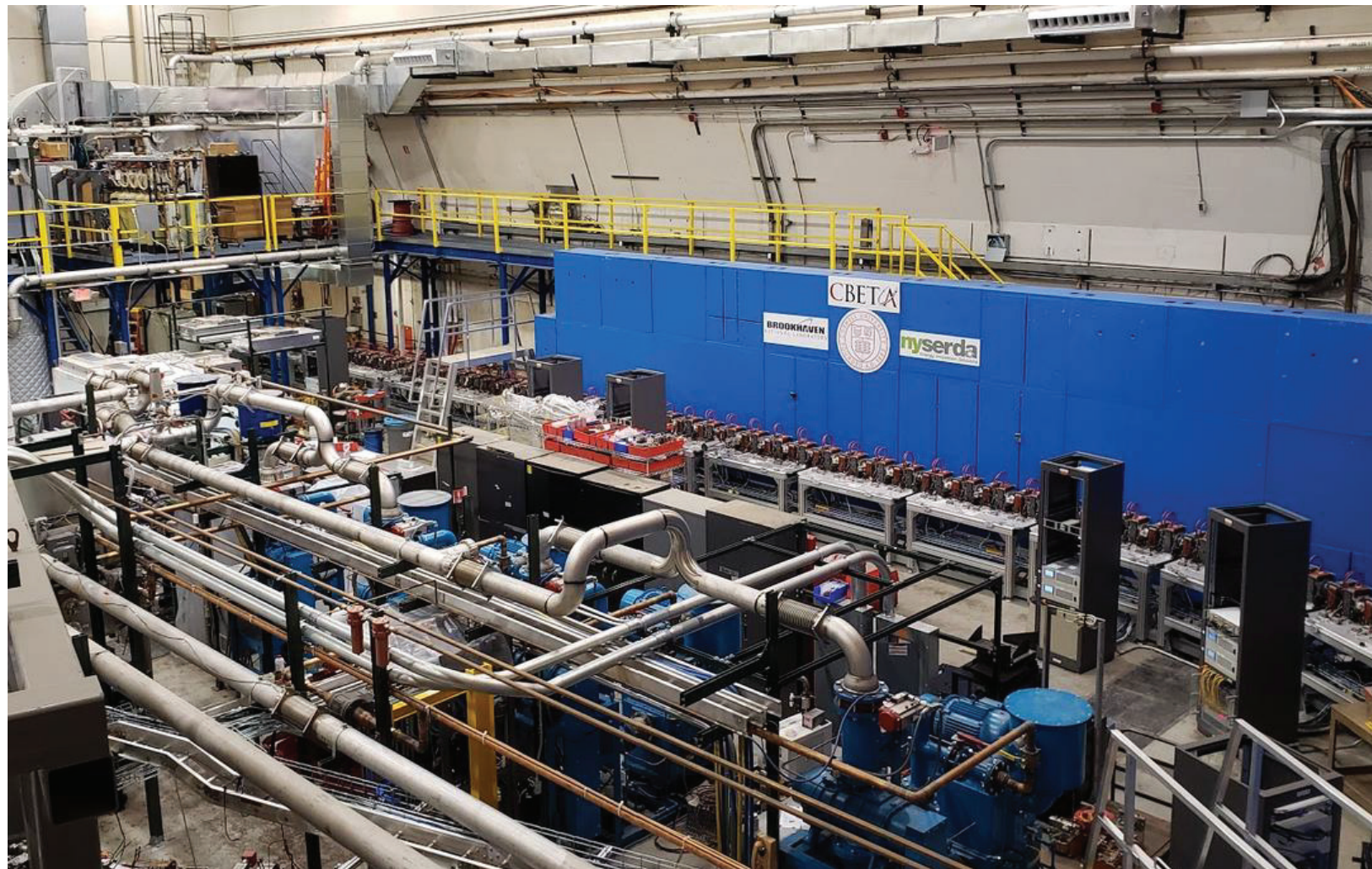




Construction



Complete!



- South side of accelerator enclosure
- FFA arc installed

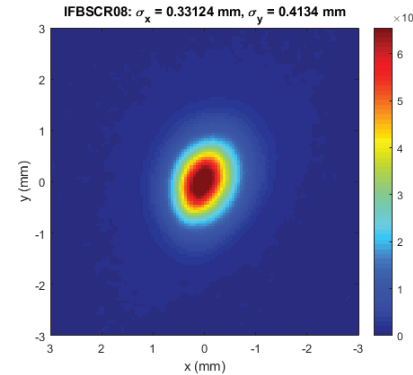
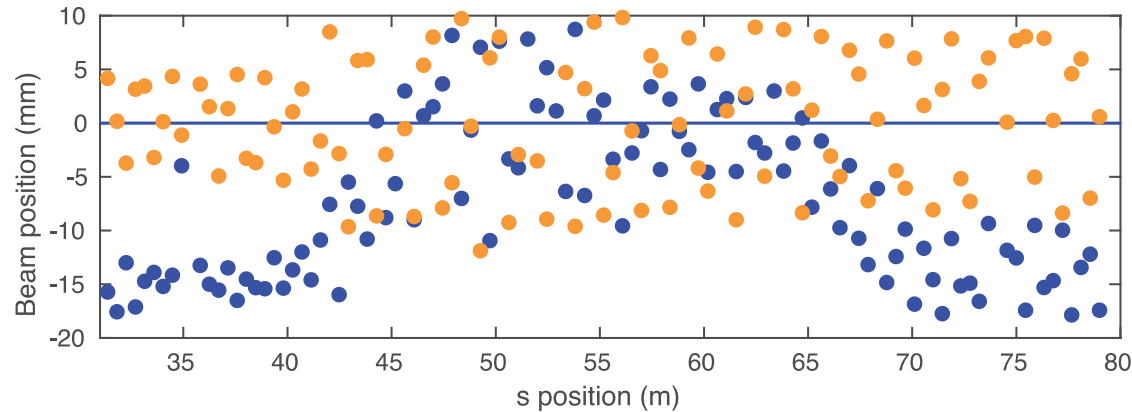


Initial Commissioning



- Commissioning began mid-March
 - Gun processing
 - Injector setup and tuning to desired optics
 - MLC and cryogenics
- Calibration and placement of internal radiation monitors
 - Critical because the permanent magnets are made of material susceptible to radiation damage
- Debug new BPM system

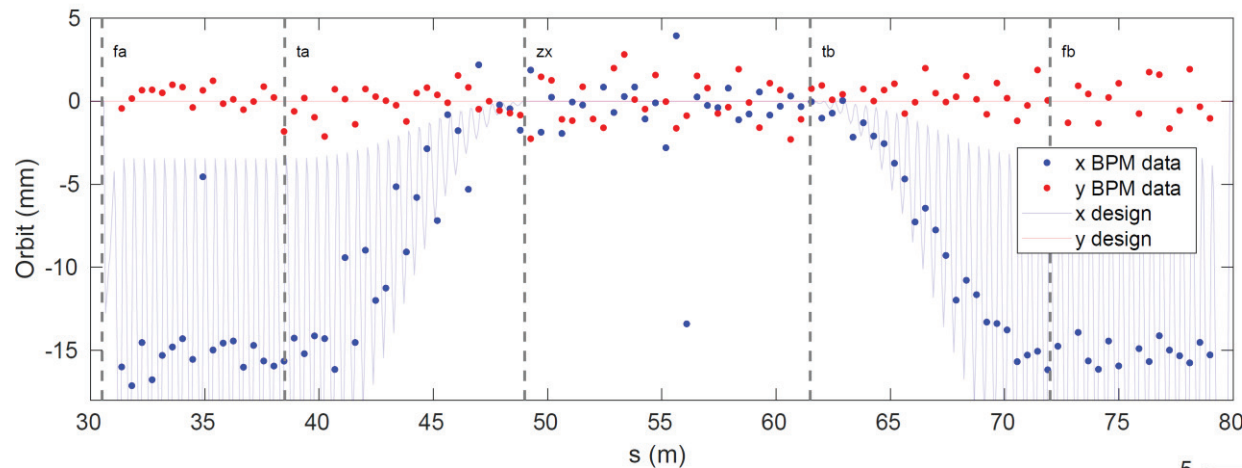
- On May 7, we put a 42 MeV beam through the entire permanent magnet arc (without correctors)



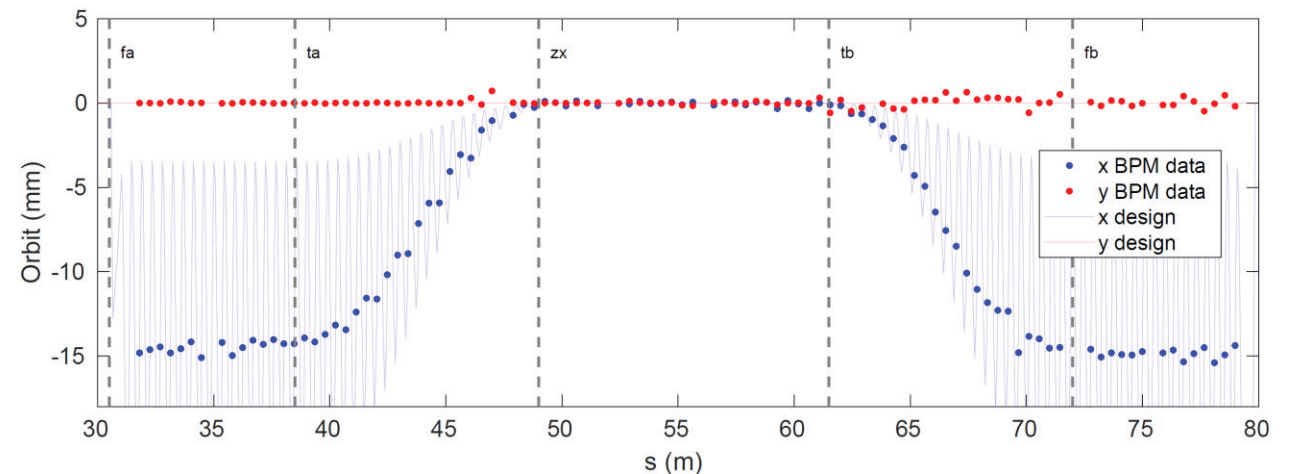
- (Top left) **Horizontal** and **vertical** beam position through FFA arc
- (Top right) Beam on viewscreen at end of the FFA arc
- (Bottom) Control room pictures from that night



- CBETA-V and virtual machine (Colwyn Gulliford)
- Orbit correction script
 - Works using response matrix from virtual machine

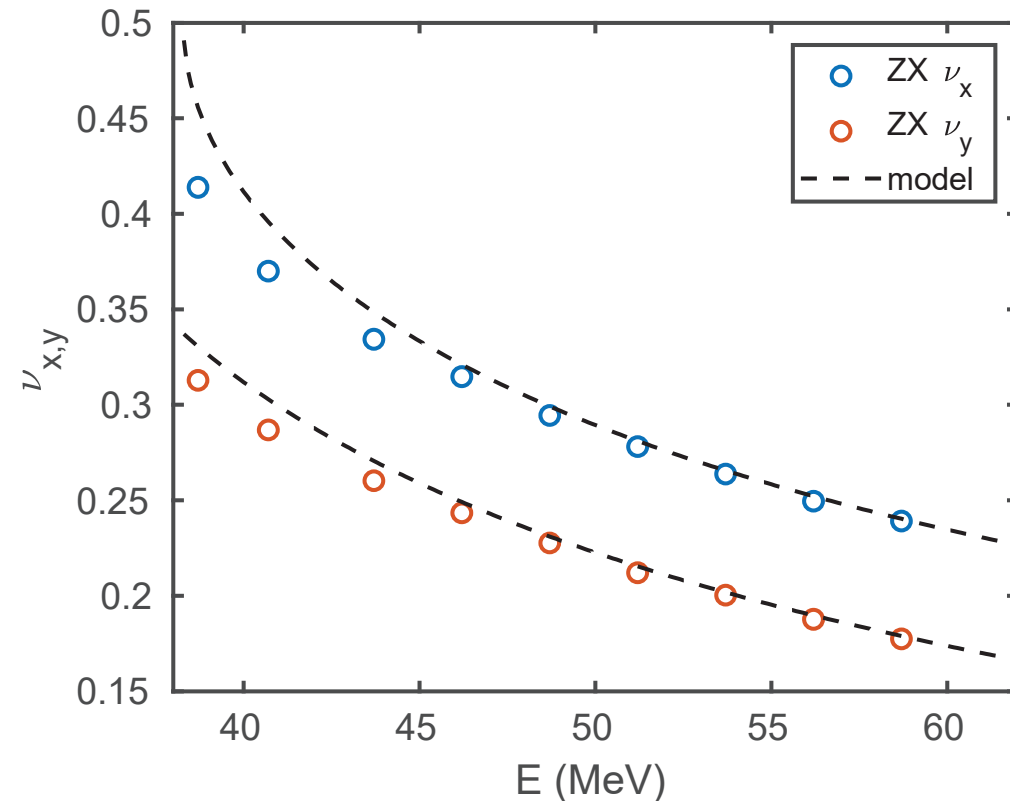
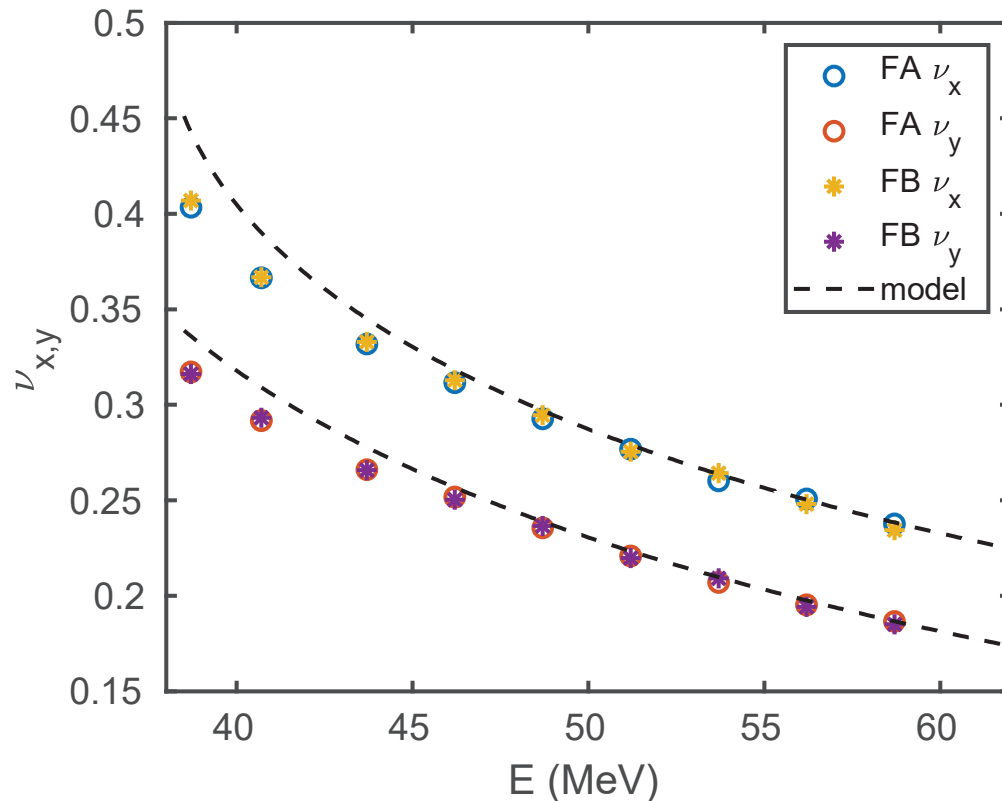


Horizontal and vertical positions on
BPMs through the FFA arc for initial
orbit (left) and corrected orbit (right)

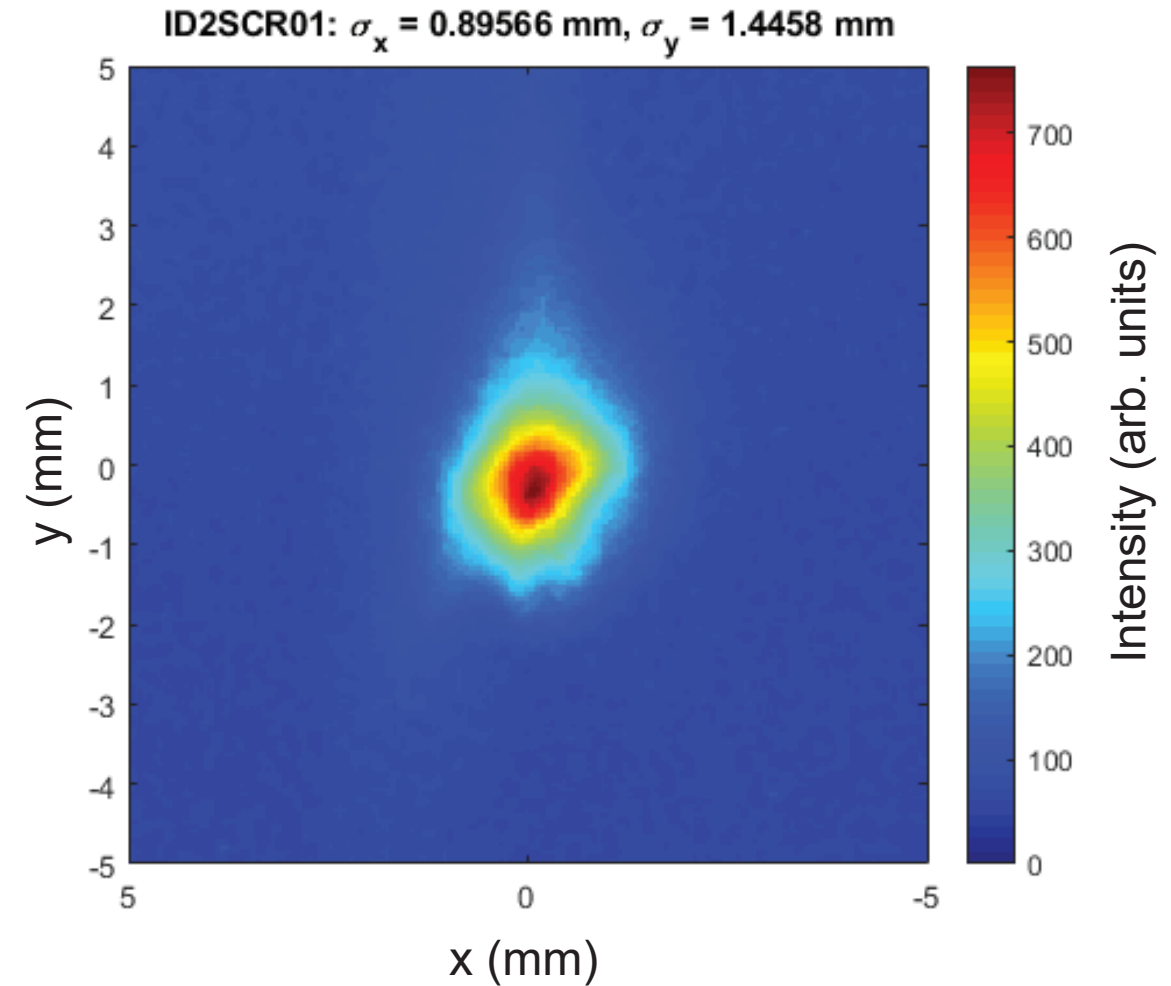


Energy Scan

- Beam energy into the arc was varied between 39 and 59 MeV
- Tunes were measured in three sections of the machine and agreed well with the model, seen below



- June 10, first recirculated beam back through the MLC into the beam stop
- Beam energy at roughly 6.1 MeV (injected energy of 6 MeV)
- Beam on first viewscreen in beam stop line, shown on right

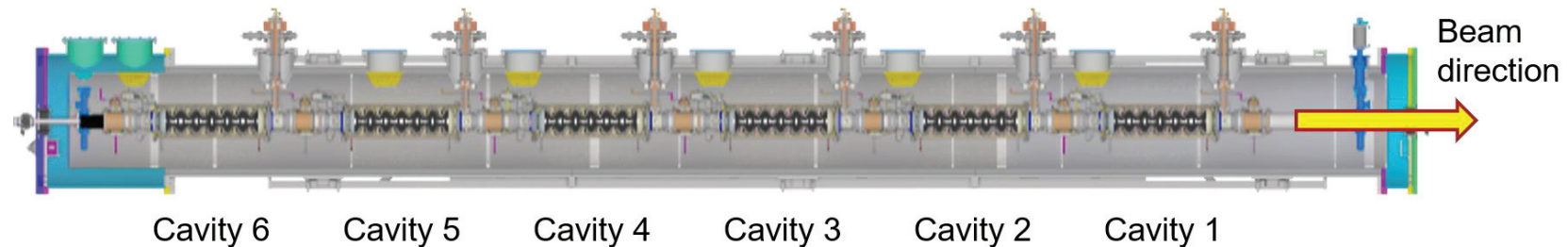
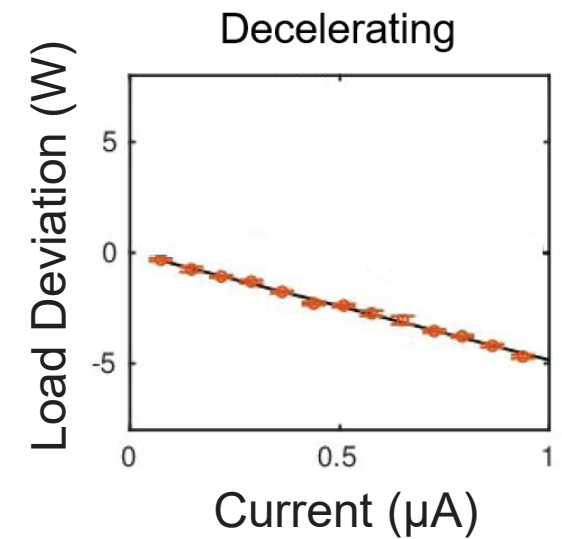
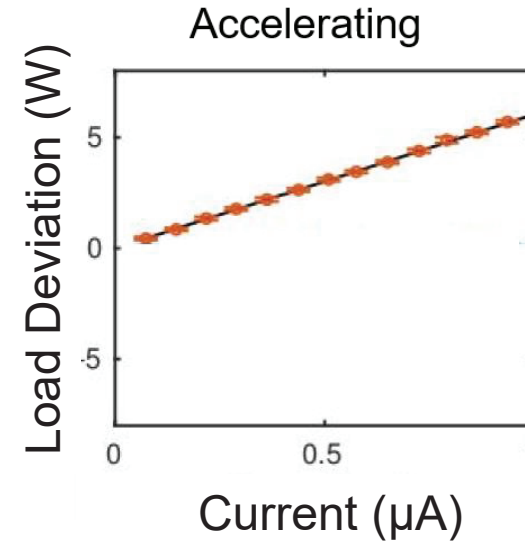




- Increased beam current to calculate transmission using RF load due to beam on MLC cavities
 - Total RF load = RF load for cavity field + RF load due to beam (rough approximation)
 - As beam current changes, the load deviation is due to the changing RF load due to beam
- First, demonstrate measurable load deviation in the MLC up to 1 μA without recirculated beam (non-ER mode)
- In perfect energy recovery mode, two conditions are satisfied:
 - There is no load deviation at any beam current
 - The linear fit of load deviation as a function of beam current is zero
- To measure energy recovery transmission, determine linear fit of load deviation as a function of current for each MLC cavity
 - The closer each linear fit slope is to zero, the better the energy recovery

Energy Recovery

- To demonstrate load deviation in MLC cavity, the each cavity was set up near typical field – but the last three cavities decelerated the beam, sending it directly into the beam stop
- MLC labeled layout shown below, typical beam loading plots for accelerating (cavities 4-6) and decelerating (cavities 1-3) shown to the right

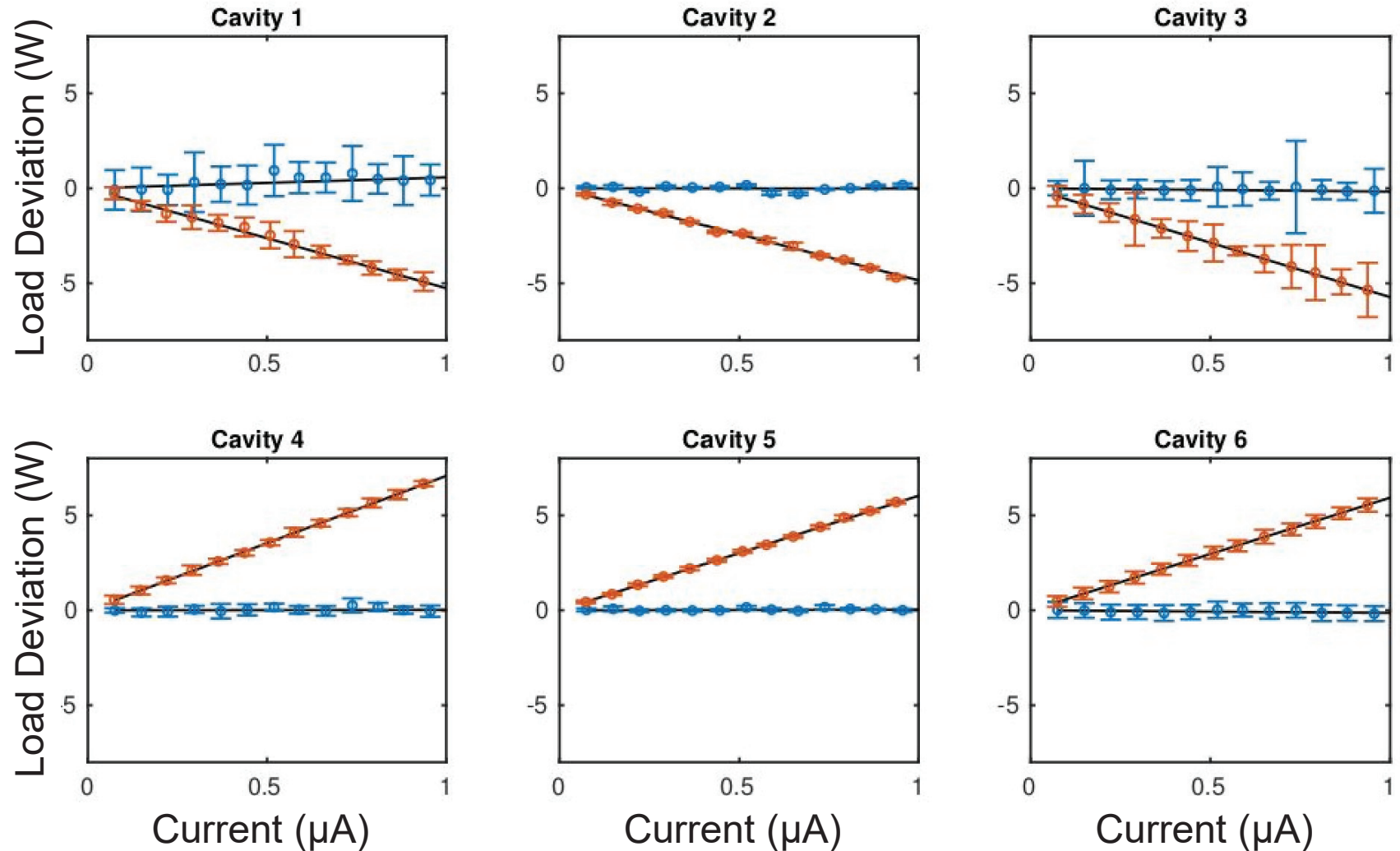




- First transmission calculation of $87.0 \pm 0.6\%$ on June 24, 2019
- Beam loading for each cavity shown on the right

ER mode

Non-ER mode

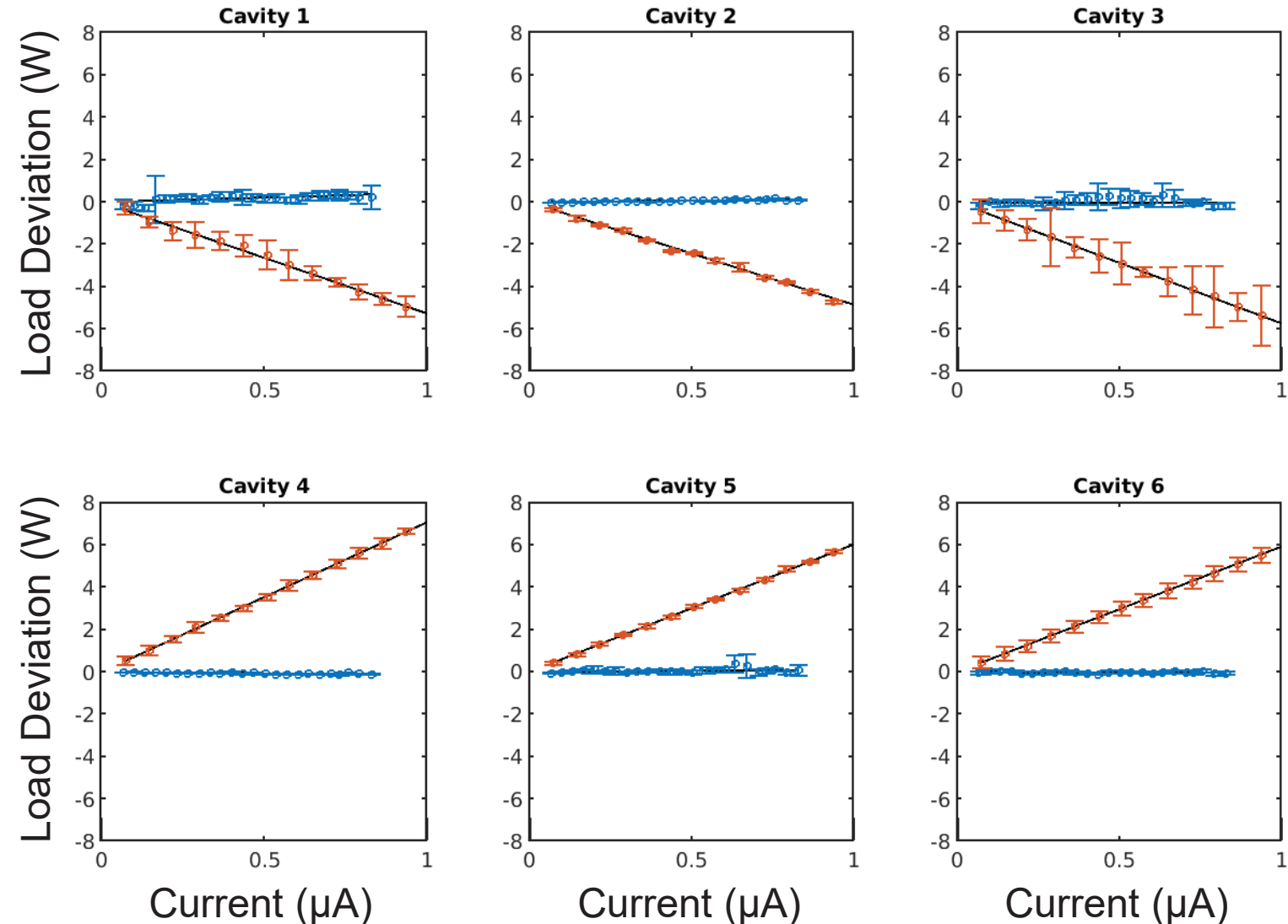




- Transmission improved to $99.3 \pm 0.2\%$ (two days later)
- Beam loading for each cavity shown on the right

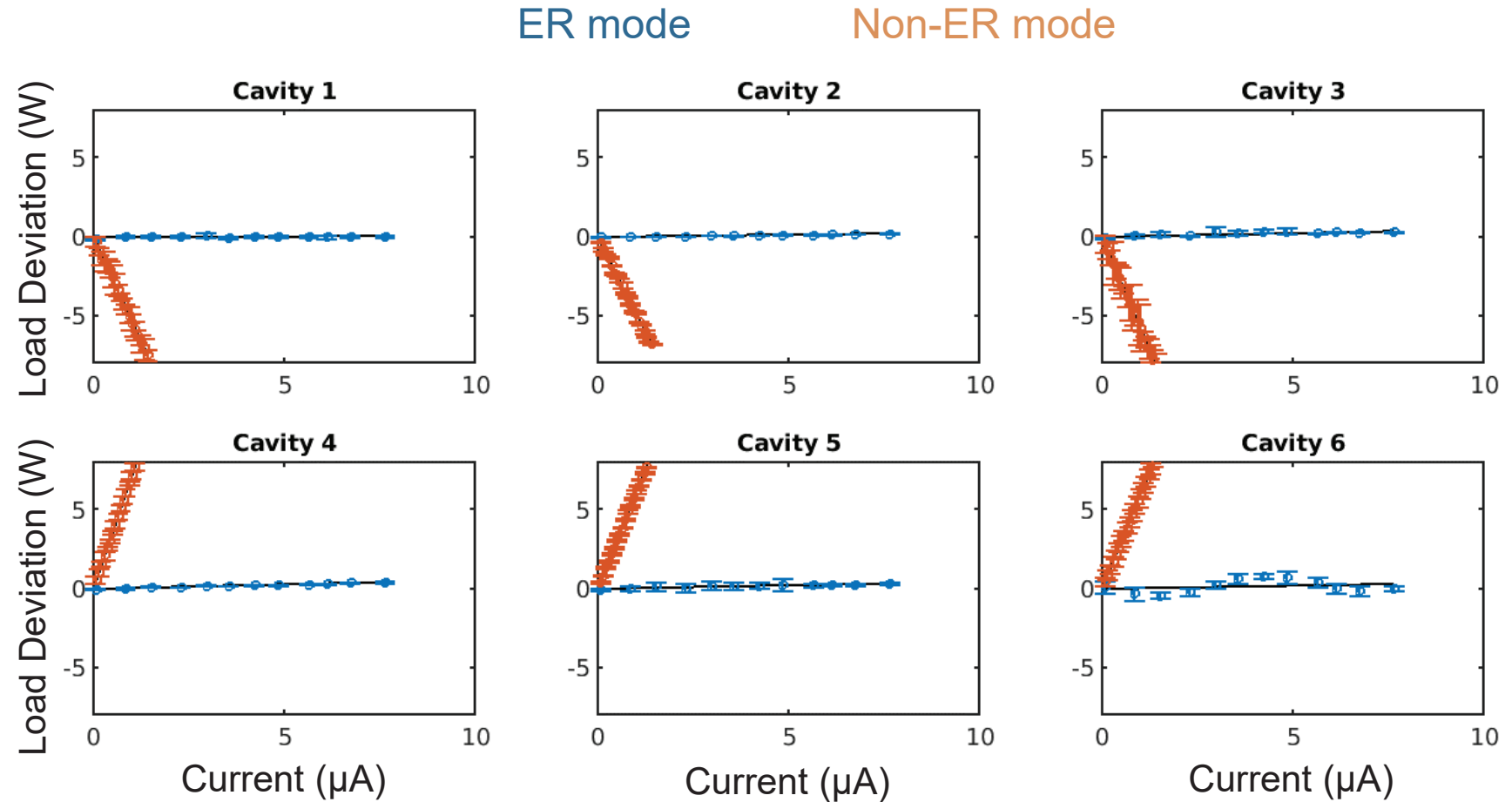
ER mode

Non-ER mode



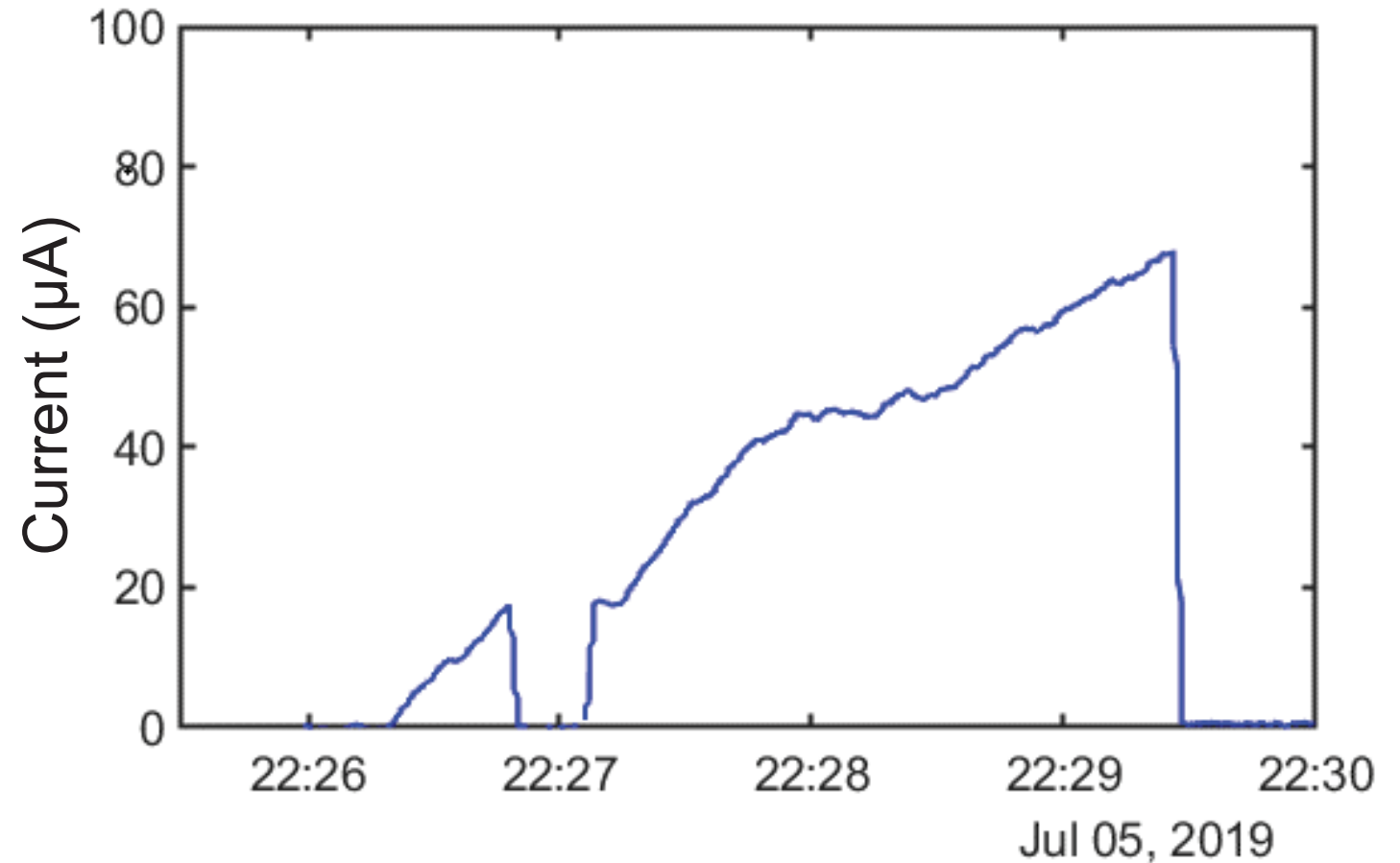


- Transmission improved to $99.6 \pm 0.1\%$
- Average current increased to $8 \mu\text{A}$
- Beam loading for each cavity shown on the right





- Decided to push for higher current – achieved 70 μA !
- Plot of the current in the gun is shown on the right

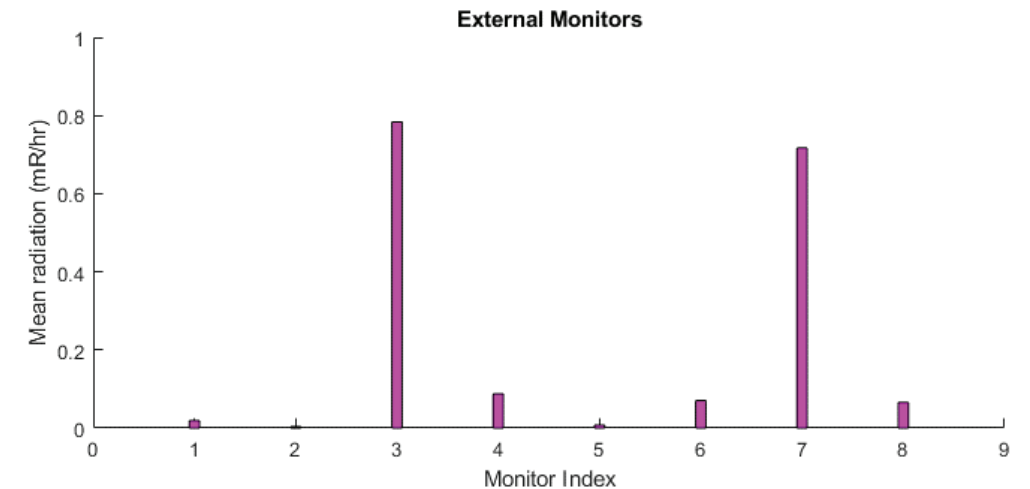
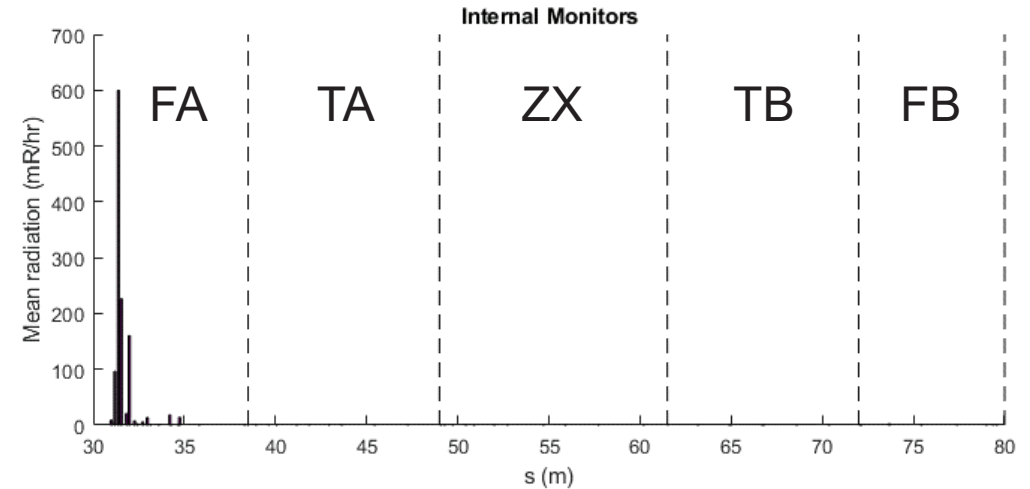




Higher Current



- Limiting factor for high average current is external radiation – not radiation in the permanent magnet arc; plots of both are shown on the right
- Currently believe radiation can be addressed by increased shielding around the beam stop





- Recently shut down for maintenance and to reconfigure as a 4-pass machine
- Commissioning of the 4-pass machine is planned to begin this fall
- After full commissioning, there are a number of potential applications, including:
 - Continued commissioning for EIC studies (including electron cooling)
 - DarkLight – an experiment to find dark matter
 - Compact Compton source for hard x-rays
 - Beam for time-resolved electron diffraction from 1-6 MeV
 - Generic ERL accelerator physics
 - and more!



Conclusion



- In less than 4 months, CBETA went from gun processing to achieving 70 μA average current, >99% transmission energy recovery, and an energy scan through the FFA arc
- Last week of beam running was spent verifying correction scripts necessary for best setup of the machine and successful 4-pass operation
- CBETA 1-pass commissioning proceeded quickly and achieved several milestones
- 4-pass commissioning to take place in the fall
- After commissioning, CBETA has numerous potential applications for physics research and accelerator R&D



Acknowledgements



- Thank you to everyone on the CBETA team, including:

N. Banerjee, A. Bartnik, I. Bazarov, J. Crittenden, L. Cultrera, J. Dobbins, C. Gulliford, G. H. Hoffstaetter (PI), W. Lou, P. Quigley, D. Sagan, K. Smolenski, V. Vesherevich, D. Widger (Cornell)

J. S. Berg, S. Brooks, R. Hulsart, R. Michnoff, S. Peggs, D. Trbojevic (PI) (BNL)

All our visiting collaborators

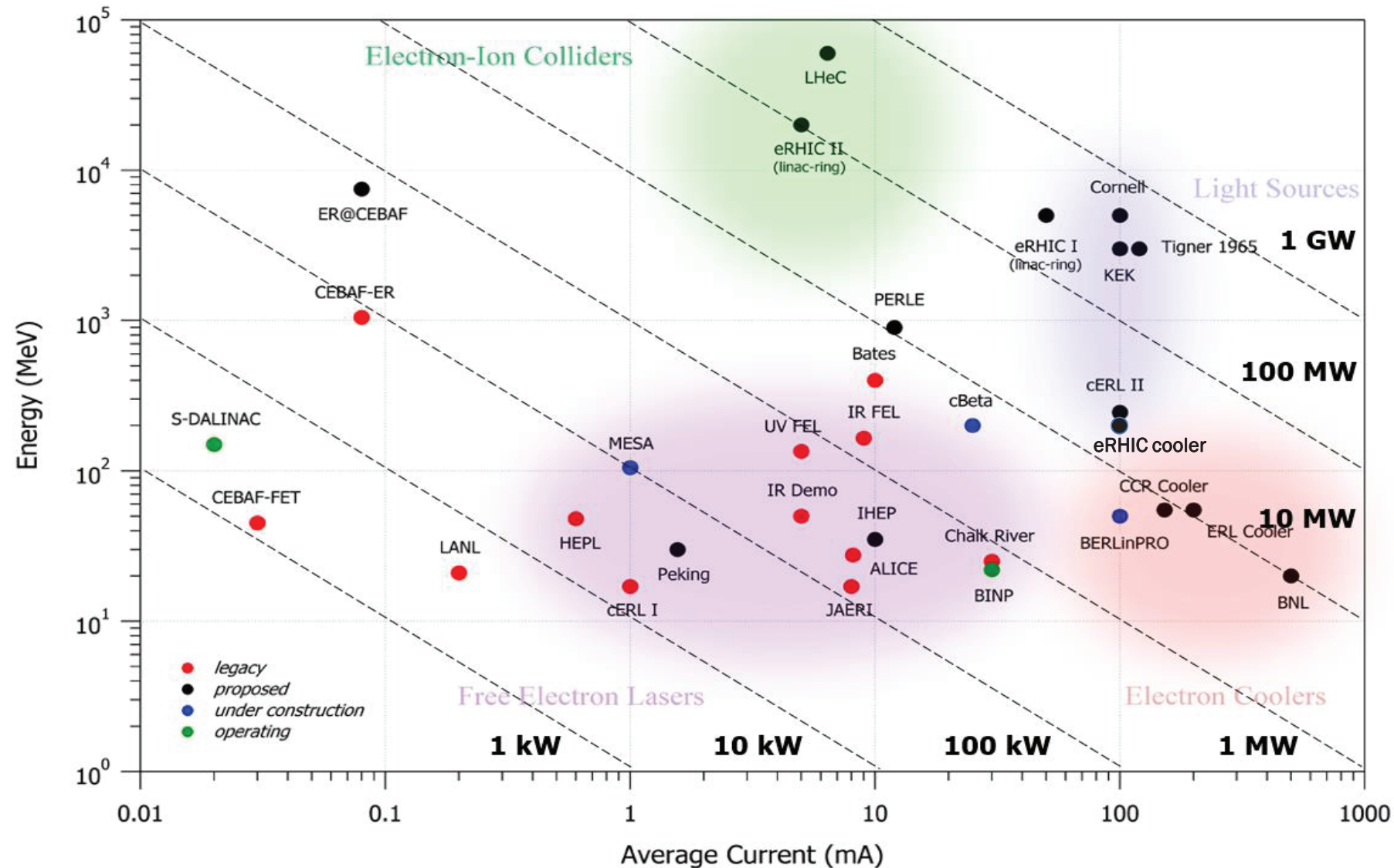
And more...

- We would like to thank CLASSE, Cornell University, Brookhaven National Lab, NSF, and NYSERDA for their funding and support in this project



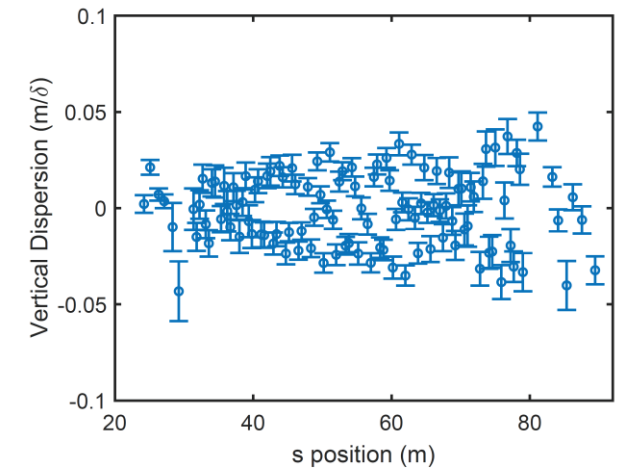
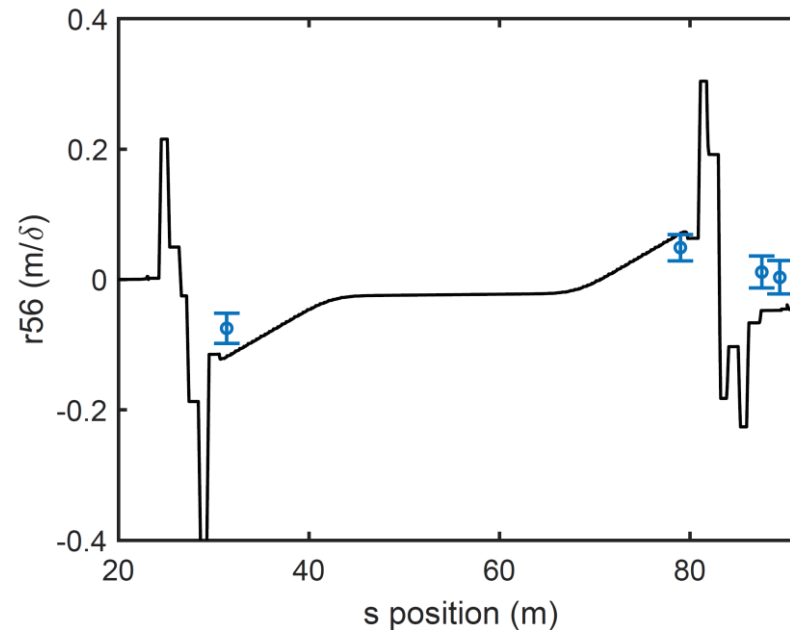
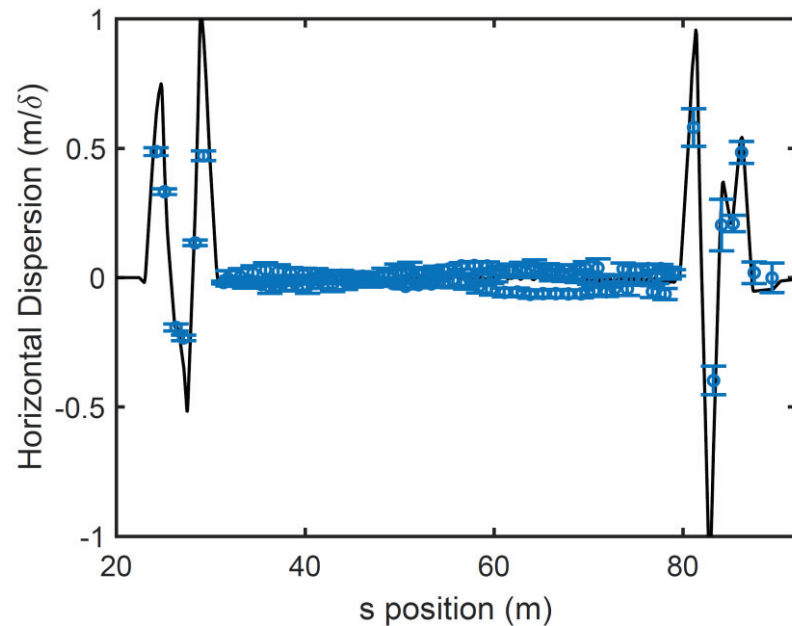
- G. Hoffstaetter *et al.*, “Energy recovery achieved in CBETA, the Cornell-BNL ERL Test Accelerator”, Internal Report CBETA-45 (06/24/2019)
- G. Hoffstaetter *et al.*, “CBETA Design Report, Cornell-BNL ERL Test Accelerator”, <https://arxiv.org/abs/1706.04245>
- C. Gulliford *et al.*, “Beam Commissioning Results from the CBETA Fractional Arc Test”, <https://arxiv.org/abs/1902.03370>
- SRF/LLRF/MLC
 - N. Banerjee *et al.*, “RF Commissioning of the CBETA Main Linac Cryomodule”, <https://srf2019.vrws.de/papers/thp023.pdf>
 - N. Banerjee *et al.*, “Active Suppression of Microphonics Detuning in High Q_L Cavities”, <https://srf2019.vrws.de/papers/weteb6.pdf>

ERL Frontier: Tennant Plot



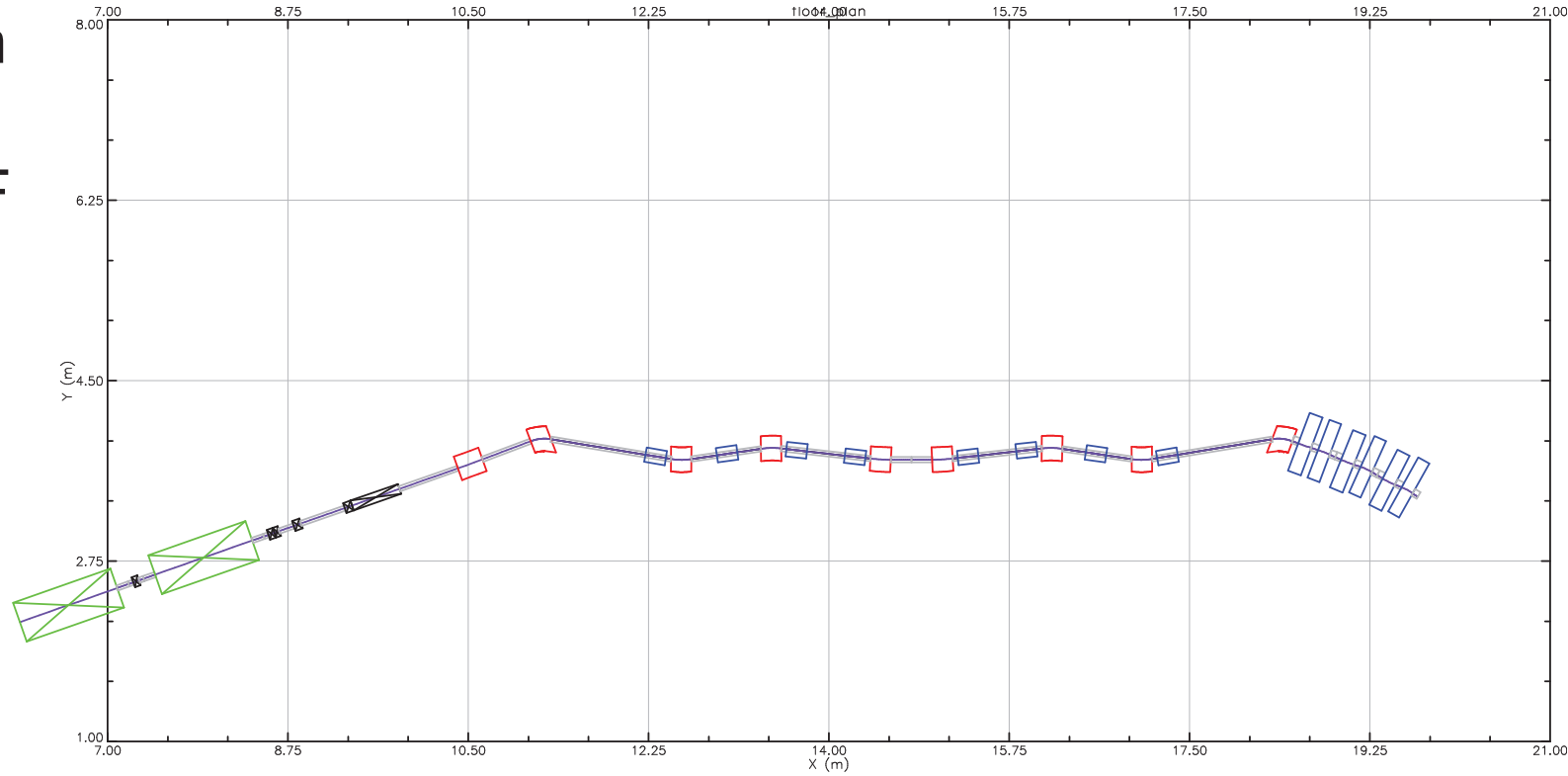


- Dispersion / R56 correction script (S1, R1)
 - Calculates response script using the real machine, determines appropriate magnet changes to achieve target values
 - Beam is stable (not moving) in the dump line



SX: 1-pass ER Configuration

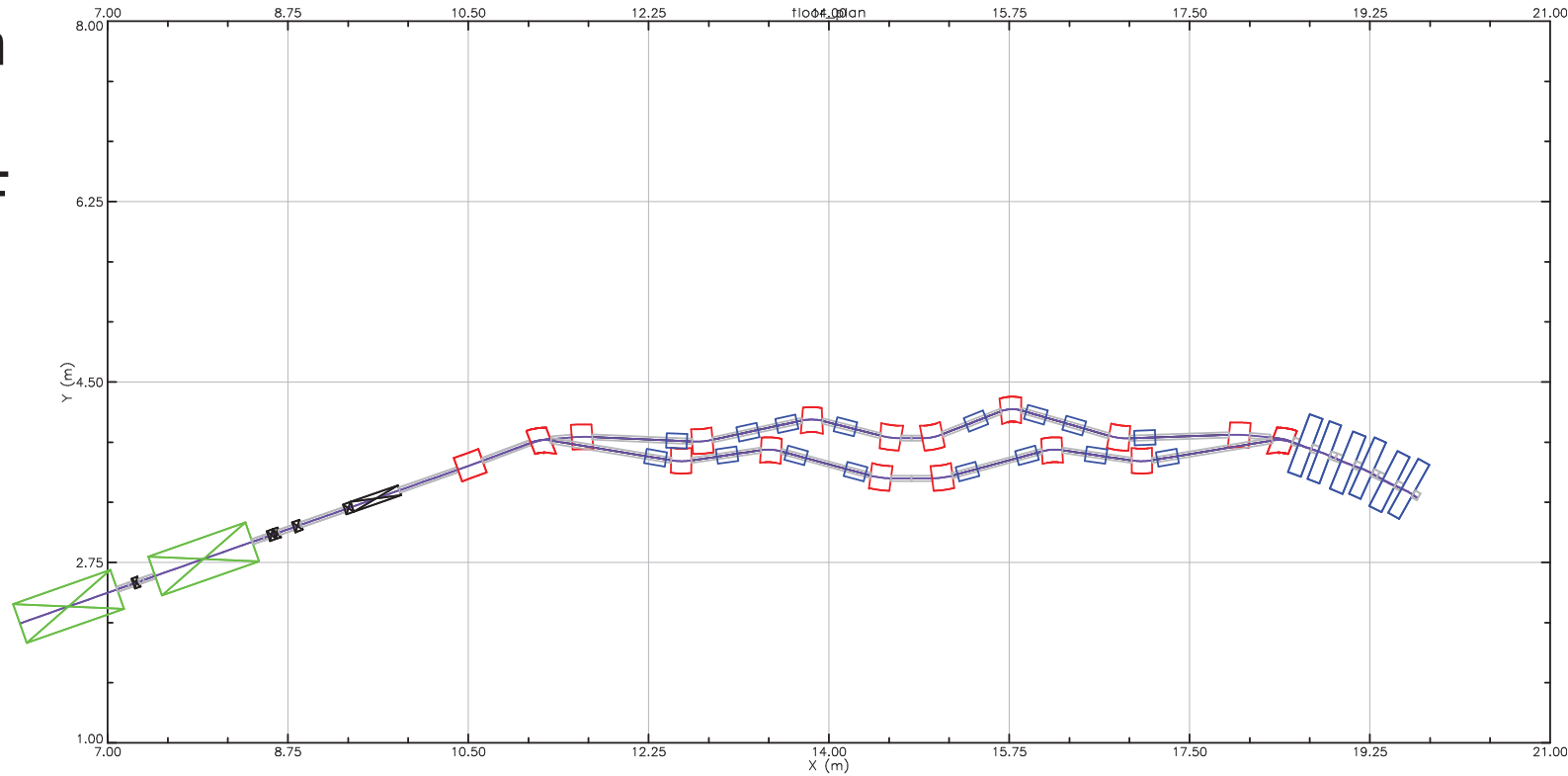
SX layout, with
path length for
each line in RF
periods



$$T_1 \cdot f_{\text{rf}} = 343 - 0.5$$

SX: 2-pass ER Configuration

SX layout, with
path length for
each line in RF
periods

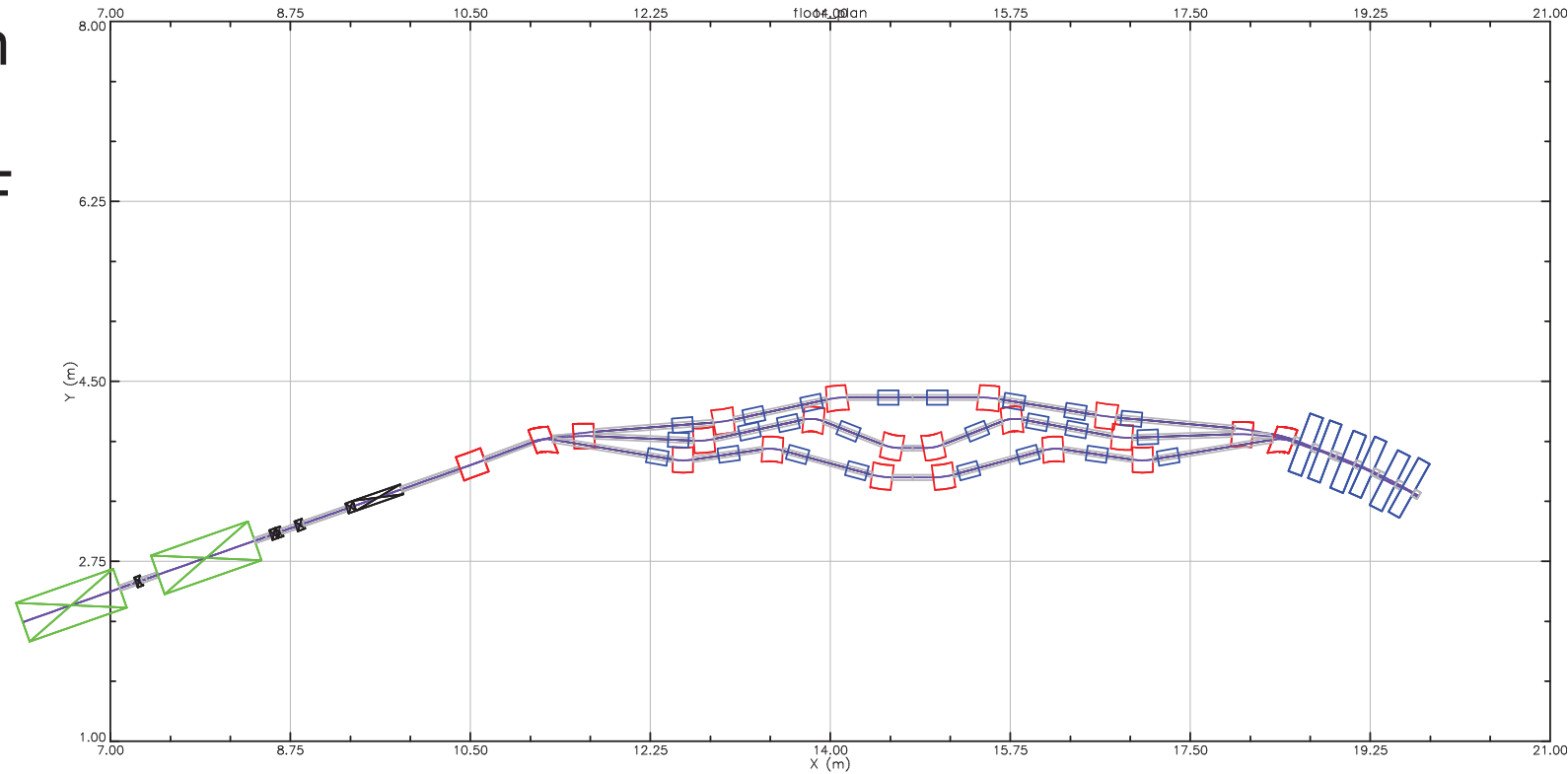


$$T_2 \cdot f_{\text{rf}} = 343 - 0.5$$

$$T_1 \cdot f_{\text{rf}} = 343$$

SX: 3-pass ER Configuration

SX layout, with
path length for
each line in RF
periods



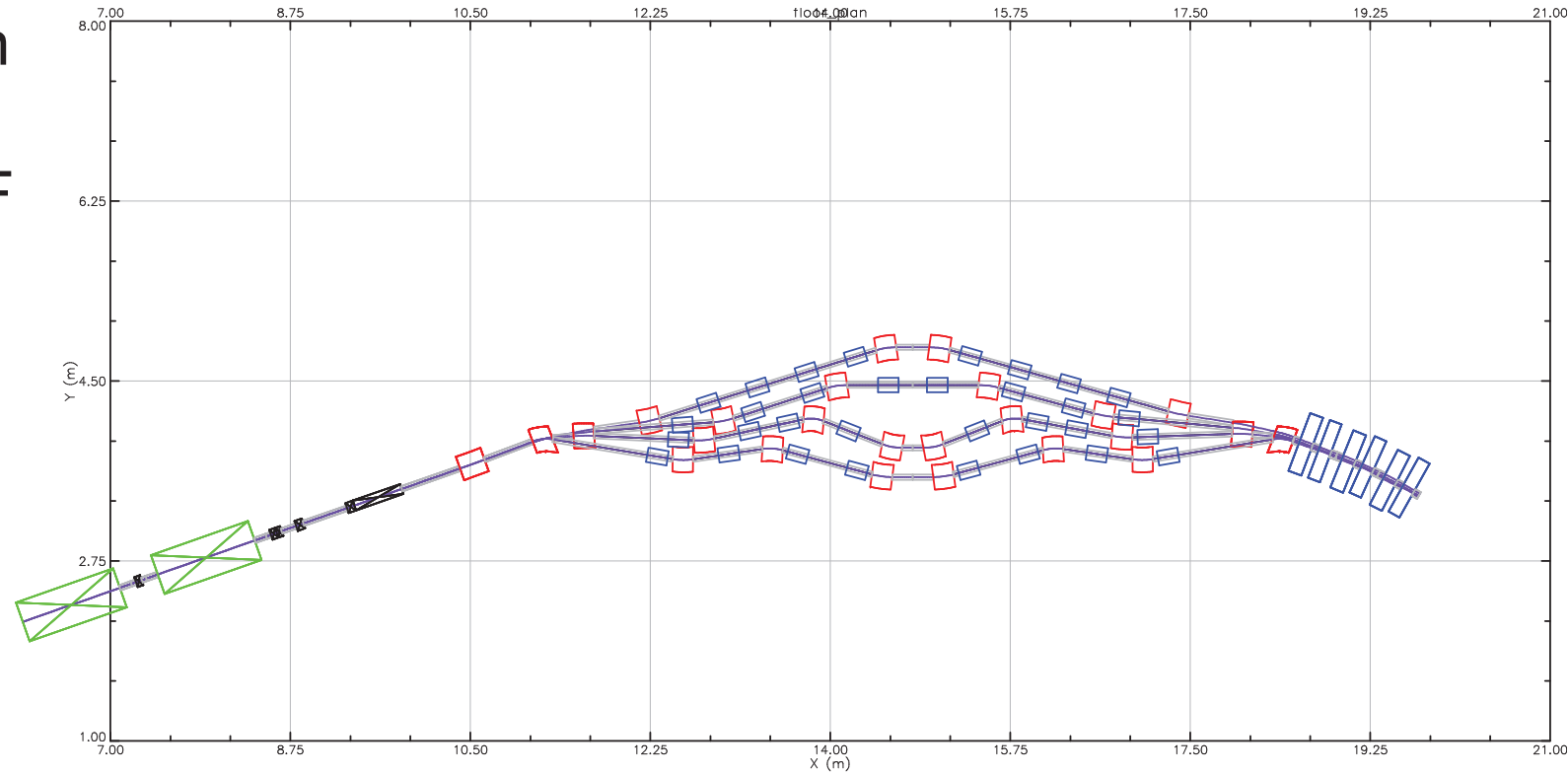
$$T_3 \cdot f_{\text{rf}} = 343 - 0.5$$

$$T_2 \cdot f_{\text{rf}} = 343$$

$$T_1 \cdot f_{\text{rf}} = 343$$

SX: 4-pass ER Configuration

SX layout, with
path length for
each line in RF
periods



$$T_4 \cdot f_{\text{rf}} = 343 + 1.5$$

$$T_3 \cdot f_{\text{rf}} = 343$$

$$T_2 \cdot f_{\text{rf}} = 343$$

$$T_1 \cdot f_{\text{rf}} = 343$$